

Data Card — Manda African Tokenizer Corpus (MATC v0.1)

Challenge 2 submission · Team Manda · TRI AI Cohort 10 · Project 3: Bridging the Tokenization Gap in African LLMs via SuperBPE

Field	Entry
Dataset name	Manda African Tokenizer Corpus (MATC), version 0.1
Purpose	Training corpus for a SuperBPE subword-and-superword tokenizer covering multiple African languages
Team	Law Adagbon — solo participant (no additional team members, no assigned mentor)
Status	Design document. The corpus is under assembly; this card describes committed design decisions, not a completed artefact.
Card version	0.1 — to be revised at Challenge 3 and at final submission

Context: what this dataset is for

Standard tokenizers are trained overwhelmingly on English. Applied to morphologically rich and agglutinative African languages, they shatter words into fragments that carry no meaning. The same sentence costs more tokens, so it costs more money, fills the context window faster, and is modelled worse. Our Challenge 1 problem statement named this as a compounding inequality: the speakers who can least afford the premium pay it every time they use an AI system in their own language.

This dataset exists to train a tokenizer that stops doing that. It is therefore unusual among the datasets in this cohort: it carries no task labels, trains no predictive model, and is never shown to a human as output. Its influence is nevertheless structural — a tokenizer sits underneath every downstream model built on it, and whichever languages are thin in this corpus will stay expensive to speak to for as long as the tokenizer is in use. Getting the composition right is the whole ethical weight of the project.

1. Input vs. output

What are users inputting? What are the outputs or labels your model will predict or generate?

Input. Plain running text: paragraphs of naturally occurring written African-language prose, unlabelled. No audio, no images, no structured records. Each training unit is one paragraph, treated independently, so no learned unit can span a document boundary.

Output. The artefact learned from this dataset is not a prediction. It is a vocabulary of at most 20,000 tokens together with the procedure that segments text into them. At inference the tokenizer turns a list of strings into a list of integer sequences and back again, under a hard constraint: decoding must reproduce the input exactly, character for character. Nothing may be normalised away, silently corrected, or lost.

What counts as success. Fewer tokens for the same text, subject to exact reconstruction. We will report this per language rather than as a single average, because an average is precisely the statistic that lets a gain on Swahili conceal a loss on Tigrinya.

2. Classification problem

What is the classification task, if any? What categories or classes are you working with, and why?

There is no classification task. Tokenizer learning is unsupervised: the algorithm counts adjacent symbol pairs and merges the frequent ones. Inventing a classification framing to fill this box would misdescribe the system, and misdescription is exactly the failure the Data Cards method is meant to catch.

One categorical decision does sit inside the dataset, however, and it deserves to be documented as if it were a label, because it carries error and it changes what the model learns:

Category	Values	Why it matters
Language	One label per document, from the language list fixed in Section 4	Determines sampling weights. A mislabelled document trains one language's vocabulary on another's text.
Script / orthography	Latin; Latin with diacritics; Ge'ez; Arabic-script (Ajami)	The evaluation pipeline transliterates non-ASCII to markers, so script predicts who is disadvantaged (Section 6).
Register	News; encyclopaedic; religious; conversational / social; literary; instructional	Register quotas are the main defence against a corpus that is 80% news and scripture.
Diacritic completeness	Proportion of tokens carrying expected diacritics	Distinguishes properly written text from the undiacritized text most people actually type.

3. Labeling

Where do your labels come from? How will you check for consistency and accuracy?

The four categories above are assigned in three passes, in decreasing order of trust:

- **Inherited metadata.** Where a source corpus already declares the language, we take its declaration and record that we did.
- **Automatic language identification** for anything without reliable metadata. We treat its output as provisional. Automatic identification is weakest exactly where our project cares most: between closely related languages, on short passages, and on code-switched text, which it tends to resolve toward the higher-resource neighbour.
- **Native-speaker verification** on a stratified sample — a fixed number of documents per language, drawn across registers, checked by a speaker of that language.

Consistency check. A portion of the verification sample is double-checked by a second speaker. We report inter-annotator agreement (Cohen's kappa) per language rather than pooled, disagreements are adjudicated by a third speaker, and the agreement figures are published in the final card whatever they turn out to be. A language whose agreement falls below an acceptable threshold is reported as low-confidence rather than quietly retained.

What we will not do. We will not use an LLM to label language identity and present it as verification. Model-assigned labels inherit the same bias toward high-resource languages that the project exists to correct, and dressing them up as ground truth would launder that bias into our documentation.

4. Data creation

How will you collect your data? Will it come from direct sources, or will you synthesize data in low-resource settings?

No primary collection from human subjects. We are not recording interviews, running surveys, or gathering speech. MATC is assembled from existing published corpora, chosen for

licence clarity and register spread. This is a deliberate scoping decision for a 16-week project: primary collection done properly needs consent processes and community agreements we cannot responsibly compress into the timeline, and collection done improperly is worse than not collecting.

Candidate source	Character	Licence position
FLORES-200 / SIB-200	Professionally translated, broad language coverage, small	CC-BY-SA-4.0. Share-alike, so it cannot be pooled into a redistributed corpus with non-commercial material.
MAFAND-MT (Masakhane)	News register, African-led curation	Per-dataset licences; each checked individually before inclusion.
Wikipedia (African-language editions)	Encyclopaedic register, uneven depth by language	CC-BY-SA. Same share-alike constraint.
ParaCrawl bonus sets (en-sw, en-so)	Web-mined parallel text	CC0. Compatible with everything.
WURA	Large; audited mC4 plus crawled news; strongest coverage of the 16 languages it spans	Provenance risk. Recent licence-compatibility work notes that redistributing WURA text means redistributing copyrighted web material under terms the copyright holders never granted. We use it under the no-redistribution design below and flag it explicitly.

Synthesis. We do not generate synthetic text to pad low-resource languages. For a tokenizer this would be actively harmful: merges are learned from frequency, so synthetic text would teach the vocabulary the statistical habits of whichever model produced it, and we would then report an efficiency gain on a language whose real writing we had never measured. Where a language has too little genuine text, the honest output is a documented gap, not a fabricated corpus.

5. Web collection ethics

Do you plan to source data from the web? Which platforms? What tools? Have you addressed consent, copyright, and privacy?

Yes, indirectly, and this is the sharpest ethical issue in the project. We do not crawl the web ourselves in v0.1. But every large corpus above is ultimately web-derived, so we inherit its provenance problems whether or not we performed the crawl. Saying 'we used a public dataset' does not discharge the responsibility; it relocates it.

Copyright and licence compatibility

Open licences are not automatically compatible with one another. Share-alike and non-commercial terms conflict, so a corpus pooling both cannot be redistributed under either. Our design decision follows from this:

- **We do not redistribute the text.** What we publish is the derived tokenizer vocabulary — roughly 20,000 short strings and their merge statistics — plus a manifest naming every source, its version, its licence and its sampling weight, and the scripts that reconstruct our corpus from those sources.
- Anyone can reproduce our corpus exactly by fetching the sources themselves under each source's own terms. Reproducibility is preserved without us becoming an unlicensed redistributor.

— Each source's licence is recorded in the manifest before that source is used, not after.
Sources whose terms we cannot establish are excluded.

Consent

We state this plainly rather than claiming a consent we do not have: nobody whose writing sits in a web crawl agreed to have it used for language-model tokenization. Licensing permits reuse; it is not consent, and the two are routinely conflated in dataset documentation. Our mitigations are partial and we name them as partial — we prefer sources curated by African institutions and researchers, who are accountable to the communities involved in a way an anonymous crawl is not; and we operate a documented removal route (Section 7).

Privacy

Web text contains personal information: names, phone numbers, addresses, email addresses. For most models this is a training-data risk. For a tokenizer it is a more literal one. Merges are frequency-driven, so a string that recurs often enough in the corpus can become a single token in the shipped vocabulary — a phone number repeated across scraped classified ads could end up permanently embedded in the artefact, visible to anyone who opens it.

This risk has an unusually cheap and complete mitigation, which we commit to: **the vocabulary is small enough to read**. Twenty thousand short strings can be reviewed in full by a human. We run automated pattern detection over the source text for phone numbers, emails and long digit strings, and then inspect the entire final vocabulary before release, removing any token that encodes identifying information and retraining. Very few datasets permit a complete audit of the shipped artefact; ours does, so declining to do it would be indefensible.

6. Bias and representation

Who might be left out of your dataset? How will you ensure it reflects the diversity of the community and includes underrepresented voices?

Four exclusions, in rough order of how much they worry us.

6.1 Scripts that the pipeline flattens

The competition's preprocessing reduces all text to ASCII, rendering non-ASCII characters as transliterations and bracketed Unicode markers. The effect is not evenly distributed. Latin-script languages pass through nearly intact. Languages written in Ge'ez (Amharic, Tigrinya) or in Arabic-script Ajami (Hausa, Wolof, Kanuri) are converted wholesale into verbose marker sequences — so the very languages already worst served by existing tokenizers are the ones our evaluation setting represents least faithfully. We cannot change the evaluation pipeline. We can refuse to hide the effect: our corpus retains these languages in original script, we report their results separately, and we do not claim the headline efficiency figure applies to them.

6.2 Diacritics, and the words that collapse without them

Yorùbá marks tone on all seven vowels and both syllabic nasals, plus underdots on three characters. These are not decoration. Published work on diacritic restoration finds that around 85% of words in a Yorùbá corpus carry diacritics, and roughly a third of undiacritized word types correspond to two or more distinct diacritized words. Strip the marks and *gbà* (spread), *gba* (accept) and *gbá* (hit) become one string; *ìlù* (drum), *ilu* (opener) and *ilú* (town) become another.

A tokenizer trained on flattened text learns merges over a vocabulary in which distinct words have been fused. It will look efficient, because collapsing distinctions always looks efficient by a token count. Our response: MATC holds both diacritized and undiacritized text as separately labelled strata, we measure fertility on both, and the card states that efficiency gains measured on ASCII-flattened Yorùbá are gains on a degraded representation of the language.

6.3 Register, and who writes in it

Openly licensed African-language text is dominated by news, encyclopaedic articles and religious translation — historically, scriptural text was the bulk of what existed digitally for many of these languages. Each skews toward formal, edited, written-to-be-published prose. Missing: conversation, market and trading talk, oral narrative in transcription, and the heavily code-switched writing that is how a great many people actually type, Naija Pidgin and English-Yorùbá or English-Swahili mixing included. A tokenizer tuned on newsprint will fragment a WhatsApp message.

Mitigation: explicit per-register quotas rather than taking whatever volume each source happens to offer; a cap on the share any single source or register may contribute per language; and an evaluation slice weighted deliberately toward the thin registers, so that the number we watch during development is the one that improves for underrepresented writing.

6.4 Languages, dialects and speakers with no digital text at all

Africa has well over a thousand languages. A corpus of this kind reaches a few dozen. Within each of those, standard and prestige varieties are over-represented, and the online authorship these corpora draw on skews urban, younger, male and formally educated. We handle language imbalance with capped near-uniform sampling rather than proportional sampling, so a high-resource language cannot dominate merge statistics purely by volume. We handle the rest by not overstating our reach: the card names the languages included and says nothing about the ones it does not cover.

7. Values and community alignment

How does your dataset design reflect the values you identified in Challenge 1? How will your community benefit, and how will you ensure accountability to them?

Value (Challenge 1)	How the dataset design carries it
Equity	Capped near-uniform language sampling instead of proportional sampling, so volume does not decide whose words become tokens. Per-language results reported separately, because an aggregate hides who lost.
Cultural respect	Diacritics and non-Latin scripts preserved in the corpus and their degradation by the evaluation pipeline documented rather than absorbed. No synthetic text standing in for a real language.
Inclusion	Register quotas that make room for conversational and code-switched writing; native-speaker verification of language labels; no LLM-assigned labels presented as ground truth.
Accountability	A published source manifest with licences and weights; a fully human-readable released vocabulary; a named removal route; and stated limits, including the ones that make our own results look weaker.

Benefit to the community

The direct benefit is cost and capability. A tokenizer that fragments less makes every downstream system cheaper to run and better at the language, and that saving lands with the people currently paying the premium: students, traders, teachers, and the developers building for them. Because the artefact is small and reusable, the benefit is not confined to our submission — anyone building an African-language model can adopt the vocabulary.

A second benefit is documentary. The manifest and this card record, for each language we touch, what text exists, under what licence, in what register, and where the gaps are. That

record is useful to the next team whether or not our tokenizer wins anything.

Accountability mechanisms

- **Readable release.** The vocabulary ships as plain readable strings, so a speaker of any covered language can inspect what the tokenizer learned about their language and tell us where it is wrong.
- **Removal route.** A named contact and a documented process for a community, publisher or individual to request that a source be dropped, with a commitment to retrain and reissue rather than to argue about licences.
- **Versioned, revisable card.** This card is versioned and will be revised at Challenge 3 and at final submission; superseded claims stay visible in the history rather than being edited away.
- **Reported failures.** Per-language fertility including the languages where we did badly, and the annotation agreement figures whatever they come out at.

8. Known limitations and open risks

- **The corpus is not yet built.** This card describes committed design decisions. Composition figures, per-language volumes and agreement scores are not yet available and are not asserted here.
- **Inherited provenance.** Using web-derived corpora means inheriting consent and copyright problems we did not create and cannot fully fix. The no-redistribution design limits our exposure; it does not resolve the underlying issue.
- **The evaluation setting is lossy.** ASCII flattening degrades exactly the languages the project most wants to serve. Our results describe performance under that setting, not under faithful orthography.
- **Verification is sampled, not exhaustive.** Language labels on unsampled documents rest on automatic identification, which is least reliable for related languages and code-switched text.
- **Coverage is a few dozen languages out of more than a thousand.** Success here narrows the gap for the languages included and leaves it untouched for the rest.

References

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